

EmoFlow: Appraisal-Conditioned Emotion State Dynamics via Pre-trained LLMs and Attention-based Temporal Memory

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Abstract

We propose **EmoFlow**, a computational framework for modeling human-like emotional dynamics in multi-turn dialogue. Given an initial emotion state, EmoFlow processes natural-language stimuli through three stages: (1) a pre-trained LLM-based Appraisal Layer that extracts cognitive evaluation dimensions; (2) a Bayesian updater that revises the agent’s emotion-state distribution; and (3) an Attention-based Temporal Memory module that maintains and decays historical emotion states across turns. We evaluate on EmoryNLP and MELD and compare against stateless and RNN-based baselines on emotion coherence and classification metrics.

1 Introduction

Dialogue agents powered by large language models (LLMs) produce responses that are emotionally inconsistent across conversation turns because each turn is processed independently, with no model of how the agent’s internal affective state evolves over time. Human emotion, by contrast, is continuous: it arises through cognitive *appraisal* of incoming stimuli (Scherer, 2001), persists across time, and gradually decays unless reinforced.

This gap motivates three core challenges. **Emotion elicitation**: how does a new dialogue turn change the agent’s internal emotional state? **Emotion persistence**: how do past emotional states influence the current response after multiple turns? **Emotion decay**: how does an emotional state weaken over time in the absence of reinforcing stimuli?

Prior work has addressed these challenges in isolation. Appraisal-theory-based systems (Croissant and Frister, 2024) model elicitation but ignore temporal persistence. Recurrent architectures capture sequential dependencies but lack psychologically grounded appraisal representations. No existing open system unifies appraisal-based elicitation,

Bayesian state updating, and attention-based temporal memory in a single trainable pipeline.

We present **EmoFlow**, a modular, end-to-end trainable system that jointly addresses all three challenges. Our contributions are: (i) a LoRA-adapted LLM Appraisal Layer grounded in Scherer’s Component Process Model; (ii) a Dirichlet–Categorical Bayesian updater for probabilistic emotion-state tracking; and (iii) a cross-attention Temporal Memory with learnable exponential decay, enabling coherent multi-turn emotion dynamics.

2 Problem Definition and Method Design

2.1 Problem Definition

Let $D = (u_1, u_2, \dots, u_T)$ be a multi-turn dialogue where each u_t is a natural-language utterance. At each turn t , the agent maintains an emotion-state distribution $P_t(e)$ over a discrete emotion pool $\mathcal{E}$ (Ekman’s six basic emotions or Plutchik’s eight). The goal is to produce, for every turn t : (a) a psychologically plausible emotion distribution $P_t(e)$ reflecting both the current stimulus and the agent’s emotional history, and (b) a response r_t conditioned on $P_t(e)$ that is emotionally coherent with respect to the full conversation history.

Formally, we seek a model f_θ such that:

$$P_t(e), r_t = f_\theta(u_t, \{(P_s(e), u_s)\}_{s < t}) \quad (1)$$

where the temporal context $\{(P_s(e), u_s)\}_{s < t}$ is maintained via an explicit memory store rather than implicitly through the LLM’s context window.

2.2 Method Design

EmoFlow consists of six modules arranged in a sequential pipeline:

(1) Stimulus Encoder. A pre-trained LLM (LLaMA-3-8B, frozen) encodes utterance u_t into a contextual embedding $\mathbf{h}_t \in \mathbb{R}^d$.

(2) Appraisal Layer. A lightweight LoRA adapter (Hu et al., 2022) (rank $r = 16$, $\approx 13\text{M}$ trainable parameters) projects $\mathbf{h}_t$ onto a five-dimensional appraisal vector $\mathbf{a}_t = [a_{\text{nov}}, a_{\text{ple}}, a_{\text{goal}}, a_{\text{cope}}, a_{\text{norm}}]$ following Scherer’s Component Process Model (CPM) (Scherer, 2001). This layer is trained on DailyDialog with weak appraisal supervision derived from emotion labels.

(3) Bayesian Emotion Updater. The agent maintains a categorical distribution $P_t(e)$ over $\mathcal{E}$ with a Dirichlet prior. Given new appraisal vector $\mathbf{a}_t$ and history prior $P(e \mid \text{hist})$ from the Temporal Memory, we update via:

$$P_t(e \mid \mathbf{a}_t, \text{hist}) \propto P(\mathbf{a}_t \mid e) \cdot P(e \mid \text{hist}) \quad (2)$$

where $P(\mathbf{a}_t \mid e)$ is parameterized by a learned linear mapping from appraisal space to emotion likelihood. This design keeps the emotion system probabilistic and uncertainty-aware.

(4) Attention-based Temporal Memory. A memory bank $\mathcal{M}_t = \{(\mathbf{v}_s, \mathbf{k}_s, \tau_s)\}_{s < t}$ stores (emotion vector, appraisal context, turn index) tuples for all past turns. When processing turn t , a cross-attention query $\mathbf{q}_t$ derived from $\mathbf{a}_t$ attends over all stored keys, with attention weights modulated by learnable exponential time decay:

$$w_i = \text{softmax}\left(\frac{\mathbf{q}_t \mathbf{k}_i^\top}{\sqrt{d}} \cdot \exp(-\lambda \cdot \Delta t_i)\right) \quad (3)$$

where $\Delta t_i = t - \tau_i$ is the elapsed turns since memory i was written and $\lambda > 0$ is a learnable decay rate. The history prior is $P(e \mid \text{hist}) = \sum_i w_i \mathbf{v}_i$. This mechanism allows the system to model the gradual fading of past emotional states and the heightened influence of recent stimuli.

(5) Emotion Output. A softmax layer over $\mathcal{E}$ produces $P_t(e)$, the current emotion distribution.

(6) Response Generator. The current emotion distribution $P_t(e)$ is prepended as a soft prompt to a frozen LLM (LLaMA-3-8B) to generate an emotionally conditioned response r_t , following a pre-trained-then-adapt paradigm.

Recent mechanistic interpretability work (Tak et al., 2025) demonstrates that mid-layer attention heads in LLMs encode psychologically plausible appraisal concepts, providing empirical grounding for our Appraisal Layer design. Furthermore, Zhao et al. (2024) show that injected emotion states systematically shift LLM decision-making, motivating our soft-prompt conditioning approach.

3 Related Work

Appraisal-based Emotion Modeling. Croissant and Frister (2024) introduced Chain-of-Emotion, which combines LLMs with cognitive appraisal theory to drive affective behavior in game agents. While this work establishes that appraisal dimensions can be extracted via LLM prompting, it does not model temporal emotion persistence or decay, and lacks a principled probabilistic update mechanism. EmoFlow extends this direction by integrating a trainable Appraisal Layer with a Bayesian updater and attention-based memory.

Mechanistic Interpretability of Affective Representations. Tak et al. (2025) (ACL 2025) demonstrate through causal intervention experiments that specific mid-layer attention heads in pretrained LLMs encode Scherer’s appraisal dimensions—novelty, goal relevance, and coping potential—as distinct, linearly separable directions in activation space. This provides mechanistic justification for our design choice to extract appraisal vectors via a linear projection adapter atop a frozen LLM rather than prompting alone, as the relevant information is already encoded in intermediate representations.

Emotion States and LLM Decision-Making. Zhao et al. (2024) (NeurIPS 2024) conduct systematic experiments showing that conditioning LLMs on explicit emotion states via soft prompts significantly alters their downstream behavior, including in reasoning and multi-step decision tasks. Their findings validate our Response Generator design, where the current emotion distribution $P_t(e)$ is used to steer response generation. Crucially, they also show that the effect is graded—stronger emotion conditioning produces larger behavioral shifts—motivating our use of a full distribution rather than a one-hot label.

Multi-turn Emotion Modeling in Conversation. Tu et al. (2024) (EMNLP 2024 Findings) propose a graph-based architecture for multimodal ERC that integrates commonsense, speaker, and contextual knowledge via interactive graph convolution. While their model achieves strong cross-modal fusion, it relies on a static graph constructed per dialogue and does not model continuous temporal decay of past emotional states. EmoFlow differs in two key respects: (i) we replace the graph structure with a cross-attention Temporal Memory that supports dynamic, turn-indexed weighting; and (ii)

we introduce a Bayesian updater that explicitly maintains uncertainty over the emotion distribution across turns. More broadly, prior recurrent approaches to ERC impose fixed memory horizons and do not model time-decay, while continuous-time affective models operate over pre-segmented labels rather than jointly learning appraisal and memory end-to-end. EmoFlow is the first pipeline to unify all three components—appraisal extraction, Bayesian state updating, and attention-based temporal memory—in a single trainable framework.

4 Evaluation Plan and Resources

4.1 Datasets

EmoryNLP (Zahiri and Choi, 2018) is a TV-show dialogue corpus (Friends) with per-utterance Plutchik emotion labels across 12,606 utterances in 897 dialogues. It is our primary evaluation benchmark for multi-turn emotion tracking.

MELD (Poria et al., 2019) is a multimodal multi-party conversation dataset also derived from Friends, containing 13,708 utterances with Ekman emotion labels and rich emotion transition sequences. We use the text modality only for fair comparison with text-only baselines.

DailyDialog (Li et al., 2017) provides 13,118 open-domain dialogues with Ekman emotion annotations. We use this corpus for appraisal layer pre-training, where emotion labels serve as weak supervision for appraisal dimension learning.

4.2 Evaluation Protocol

Metrics. We report (i) *weighted-F1* on per-turn emotion classification (standard ERC metric); (ii) *Emotion Transition Accuracy (ETA)*, measuring whether the model correctly predicts the direction of emotion shift given appraisal cues; and (iii) *Coherence Score*, a human evaluation of emotional consistency across a full dialogue on a 5-point Likert scale (conducted on 100 randomly sampled conversations from EmoryNLP test set, with two annotators).

Baselines. We compare EmoFlow against two baselines to enable controlled ablation:

- **Stateless LLM:** LLaMA-3-8B prompted with only the current utterance and no emotion history, representing the standard independent-turn paradigm.
- **RNN-based Memory:** An LSTM emotion tracker that takes appraisal vectors as input

and maintains a hidden state as emotion memory, without the attention-based decay mechanism.

All three systems use the same Stimulus Encoder and Appraisal Layer, so differences in performance isolate the contribution of the memory architecture. Baselines are trained and evaluated under identical data splits (EmoryNLP 70/10/20 train/dev/test; MELD official splits).

4.3 Estimated Computing Resources

Model sizes. The Stimulus Encoder uses LLaMA-3-8B (frozen, 8B parameters, ≈ 16 GB in bf16). The LoRA Appraisal Layer adds ≈ 13 M trainable parameters (rank 16, applied to Q/V projections). The Temporal Memory adds a lightweight cross-attention module (≈ 10 M parameters). The Response Generator reuses the frozen LLaMA-3-8B.

Training. LoRA fine-tuning of the Appraisal Layer on DailyDialog requires ≈ 4 –6 hours on one NVIDIA A100-40GB GPU. End-to-end training of the Temporal Memory and Bayesian Updater on EmoryNLP and MELD is estimated at ≈ 8 –12 hours per run on one A100. Ablation experiments (three configurations $\times$ two datasets) total ≈ 3 –4 days of A100 time.

Availability. All group members have access to the UC Davis CAES HPC cluster, which provides A100 GPU nodes. The entire experimental pipeline is estimated to complete within four weeks of GPU allocation, well within the course timeline.

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